Math

Class performance ৭ মার্ক থাকবে

Class ~~the~~ Test ৭ টা → 4 টার Best 3

① তৃতীয় সপ্তাহে একটা
4/5 তম সপ্তাহে ৫+২

formula class এর প্রথমে পড়াবে তারপর মডেল

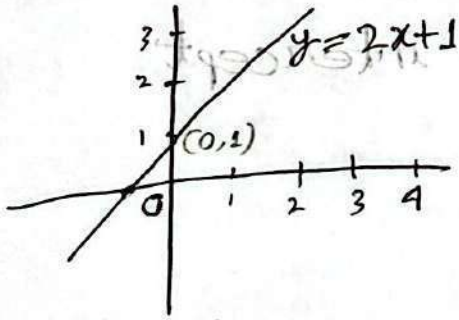
30 মার্ক আমরা class test + Presence + Performance থেকে

28 July 2026

Straight line

A straight line graph is a visual representation of a linear function. A straight line has a general equation of $y = mx + c$.

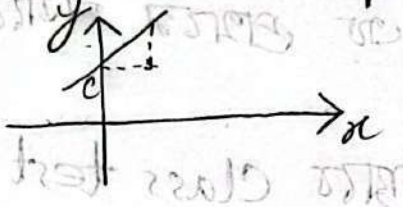
Ex:- For $y = 2x + 1$, $m = 2$, $c = 1$



Forms of straight line:-

① Slope - Intercept form:

$y = mx + c$, m is the slope and c is the y-intercept.



② Point Slope Form:

$y - y_1 = m(x - x_1)$, where m is the slope and (x_1, y_1) is a point on the line.

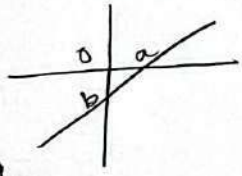
③ Standard form:

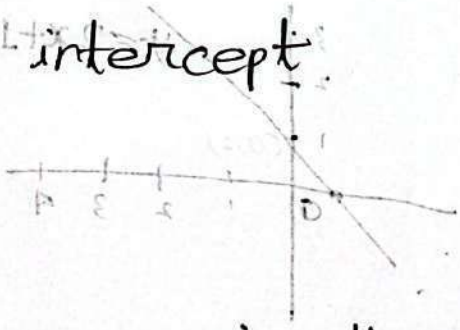
$ax + by = c$, where a, b, c are constants

$$\boxed{ax + by - c = 0}$$

④ Intercept form:

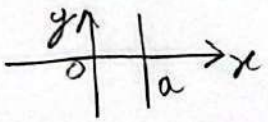
$\frac{x}{a} + \frac{y}{b} = 1$; where a is the x-intercept and b is the y-intercept.





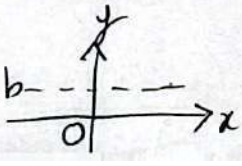
⑤ Vertical form:-

$x = a$, Vertical line with a is the x intercept



⑥ Horizontal form:

$y = b$. horizontal line with b as the y-intercept



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H.W

$$m = \text{slope} = \frac{\text{difference in } y}{\text{difference in } x}$$

Two point slope form:

The equation of a straight line passing through two points (x_1, y_1) and (x_2, y_2)

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

Ex:

Find the equation of a line passing through the point $(1, -2)$ $(4, 5)$.

Soln:-

We know that the two point slope form of straight line is

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y+2}{-2-5} = \frac{x-1}{1-4}$$

$$\Rightarrow \frac{y+2}{-7} = \frac{x-1}{-3}$$

$$\Rightarrow 3y + 6 = 7x - 7$$

$$\Rightarrow 7x - 3y = 13$$

(2,3), (5,7) বিন্দুগামী রেখার সমীকরণ

H.W. → ~~7x-8y=1~~ সমীকরণ

Parallel and Perpendicular line

(1) The equation of line parallel to $ax+by+c=0$ is $ax+by+k=0$

(2) The equation of line perpendicular to $ax+by+c=0$ is $bx-ay+k=0$

where k is an arbitrary constant to be determined by a suitable condition.

Ex: Find the equation of line passing through (-5,6) and parallel to $7x-8y=9$.

Sol^n:

The parallel line of $7x-8y=9$ is $7x-8y=k$ ... (1)

As the line is passing through (-5,6) then (1) becomes -

$$7(-5)-8\times 6=k$$

$$\Rightarrow -35-48=k$$

$$\therefore k=-83$$

Thus the required line is from (1)

$$7x-8y+83=0$$

W. H

Ex:-
Find the equation of line passing through (-5, 6) and perpendicular to $7x - 8y - 9 = 0$

Sol^n
We know that, the equation of line perpendicular to $ax + by + c = 0$ is $bx + ay + k = 0$
where k is an arbitrary constant

$a = 7, b = -8$ Thus the perpendicular line is

$$-8x - 7y + k = 0 \quad \text{①}$$

As the perpendicular line is passing through (-5, 6)

Thus, $-8(-5) - 7(6) + k = 0$

$\Rightarrow 40 - 42 + k = 0$

$\therefore k = 2$

① becomes,

$$-8x - 7y + 2 = 0$$

$$\therefore 8x + 7y - 2 = 0$$

H.W

H.W

Find the equation of lines passing through $(-1,2)$ and (a) parallal (b) Perpendicular to $3x-4y=6$.

Ans: (a) $3x-4y=-11$, (b) $-4x-3y=-2$

The equation of a line passing through the intersection of two lines:-

The equation of line passing through the two lines $ax+by+c_1=0$ and $a_1x+b_1y+c_1=0$ is

$$ax+by+c+\lambda(a_1x+b_1y+c_1)=0$$

where λ is an arbitrary constant.

Ex:- Find the equation of line passing through the intersection of two lines $2x+3y-5=0$, $x-y+2=0$

Soln: We know that,

The equation of line passing through the two lines $ax+by+c=0$ and $a_1x+b_1y+c_1=0$ is

$$ax+by+c+\lambda(a_1x+b_1y+c_1)=0$$

where λ is an arbitrary constant

So, $(2x+3y-5)+\lambda(x-y+2)=0$

$\Rightarrow 2x+3y-5+\lambda x-\lambda y+2\lambda=0$

$\Rightarrow (2+\lambda)x+(3-\lambda)y+(-5+2\lambda)=0$

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H.W

W.H

Similarly

Ex:-
$L_1 = 3x + y - 3 = 0$
$L_2 = 5x + y - 6 = 0$

(Ans) $(3+5\lambda)x + (1+\lambda)y + (3+6\lambda) = 0$

Sol^n: We know that,
The equation of line passing through the two lines $ax+by+c=0$ and $a_1x+b_1y+c_1=0$ is $ax+by+c+\lambda(a_1x+b_1y+c_1)=0$ where λ is an arbitrary constant

So, $3x+y-3+\lambda(5x+y-6)=0$
$\Rightarrow 3x+y-3+5\lambda x+\lambda y-6\lambda=0$
$\Rightarrow (3+5\lambda)x+(1+\lambda)y-(3+6\lambda)=0$

Perpendicular distance:

The length of the perpendicular distance d from a point (x_1, y_1) on the line, $ax+by+c=0$ is

$$d = \pm \frac{ax_1+by_1+c}{\sqrt{a^2+b^2}}$$

Ex:-
Find the perpendicular distance from the point $(3,4)$ to the line $5x-12y+9=0$

Sol^n
Here $a=5$, $b=-12$, $c=9$

$$D = \frac{|5.3+(-12).4+9|}{\sqrt{5^2+(-12)^2}} \Rightarrow \frac{|15-48+9|}{\sqrt{25+144}} = \frac{24}{13}$$

H.W
Find the perpendicular distance from the point $(2,-3)$ to the line $3x+4y-10=0$ [Ans: $\frac{16}{5}$]

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H.W

1. Find the equation of a line passing through the point (2,3), (5,7).

$\Rightarrow$ We know that two point slope form of straight line is

$$\frac{y-y_1}{y_1-y_2} = \frac{x-x_1}{x_1-x_2}$$

$$\Rightarrow \frac{y-3}{3-7} = \frac{x-2}{2-5}$$

$$\Rightarrow \frac{y-3}{-4} = \frac{x-2}{-3}$$

$$\Rightarrow \frac{y-3}{4} = \frac{x-2}{3}$$

$$\Rightarrow 3y-9 = 4x-8$$

$$\Rightarrow 4x-3y+1 = 0$$

2(a). Find the equation of lines passing through (-1,2) and parallal to $3x-4y=6$

$\Rightarrow$ The parallal line of $3x-4y=6$ is $3x-4y=k$ ... (1)

As the line passing through (-1,2) the (1) becomes

$$3(-1)-4\times 2 = k$$

$$\Rightarrow -3-8 = k$$

$$\Rightarrow k = -11$$

Thus the requierd line from equation (1) $\rightarrow$

$$3x-4y = -11$$

H.W

2.(b) Find the equation of lines passing through $(-1,2)$ and perpendicular to $3x-4y=6$

$\Rightarrow$
We know that, the equation of line perpendicular to $ax+by+c=0$ is $bx-ay+k=0$
where k is an arbitrary constant
$a=3$, $b=-4$ Thus the perpendular line is

$$-4x-3xy+k=0 \quad \cdots\cdots (1)$$

As the perpendicular line is passing throug $(-1,2)$
Thus $-4(-1)-3\times 2+k=0$

$$\Rightarrow 4-6+k=0$$

$$\Rightarrow k=2$$

(1) becomes

$$-4x-3y+2=0$$

3. Find the perpendicular distance from the point $(2,-3)$ to the line $3x+4y-10=0$

$\Rightarrow$
Here, $a=3$, $b=4$, $c=-10$

$$\therefore D=\frac{|3\times 2+4\times(-3)-10|}{\sqrt{3^2+4^2}}=\frac{|6-12-10|}{5}$$

$$=\frac{16}{5}$$

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~~Frequency Distribution~~

The condition of collinearity of three points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) is

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

* Check whether the points $(1,2), (3,6)$ and $(5,10)$ are collinear or not.

Sol^n :-

$$\begin{vmatrix} 1 & 2 & 1 \\ 3 & 6 & 1 \\ 5 & 10 & 1 \end{vmatrix}$$

$$= 1\begin{vmatrix} 6 & 1 \\ 10 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 1 \\ 5 & 1 \end{vmatrix} + 1\begin{vmatrix} 3 & 6 \\ 5 & 10 \end{vmatrix}$$

$$= (6-10) - 2(3-5) + 1(30-30)$$

$$= -4 - 2(-2) + 1 \cdot 0$$

$$= -4 + 4 + 0$$

$$= 0$$

Thus the points are collinear

Ex:- Check whether the points $(-1,1), (2,5), (3,6)$ are collinear or not.

Sol^n:-

$$\begin{vmatrix} -1 & 1 & 1 \\ 2 & 5 & 1 \\ 3 & 6 & 1 \end{vmatrix} = -1\begin{vmatrix} 5 & 1 \\ 6 & 1 \end{vmatrix} - 1\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix}$$

$$= -1(5-6) - 1(2-3) + 1(12-15)$$

$$= -(-1) - 1(-1) + 1(-3)$$

$$= 1 + 1 - 3$$

$$= -1$$

Thus the points are not collinear

Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represents a pair of parallel lines.

Sol^n:-

$$x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$$

$$\Rightarrow x^2 + x(6y+4) + (9y^2 + 12y - 5) = 0 \quad [\text{Quadratic in } x]$$

which is in the form $ax^2 + bx + c = 0$

where $a = 1$,

$b = 6y + 4$

$c = 9y^2 + 12 - 5$

for $ax^2 + bx + c = 0$, roots are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\therefore x = \frac{-(6y+4) \pm \sqrt{(6y+4)^2 - 4.1.(9y^2+12y-5)}}{2.1}$$

$$= \frac{(-6y-4) \pm \sqrt{(6y)^2 + 2.6y.4 + (4)^2 - (36y^2+48y-20)}}{2}$$

$$x = \frac{-6y-4 \pm 6}{2}$$

$$x = -3y - 2 \pm 3$$

$$\therefore x + 3y + 5 = 0$$

$$x + 3y - 1 = 0$$

As these two lines differ only in constant terms, so these form a pair of paralled straight lines

Prove that the equation $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$ represents a pair of ~~parallel~~ perpendicular ~~lines~~ lines

Soln $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$

$\Rightarrow -3x^2 + x(-8y - 29) + (3y^2 + 3y - 18) = 0$

Comparing with the quadratic equation

$ax^2 + bx + c = 0$

here, $a = -3$

$b = -8y - 29$

and $c = 3y^2 + 3y - 18$

Thus the roots are,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-8y-29) \pm \sqrt{(-8y-29)^2 - 4(-3)(3y^2+3y-18)}}{2(-3)}$$

$$\therefore x = \frac{8y+29 \pm \sqrt{(-8y)^2 - 2(-8y).29 + (29)^2 + 12(3y^2+3y-18)}}{-6}$$

$$= \frac{8y+29 \pm \sqrt{64y^2+464y+841+36y^2+36y-216}}{-6}$$

$$\Rightarrow x = \frac{8y+29 \pm \sqrt{100y^2+500y+625}}{-6}$$

$$\Rightarrow x = \frac{8y+29 \pm \sqrt{(10y)^2 + 2.10y.25 + (25)^2}}{-6}$$

$$\Rightarrow x = \frac{8y+29 \pm \sqrt{(10y+25)^2}}{-6}$$

$$\Rightarrow x = \frac{8y+29 \pm (10y+25)}{-6}$$

Now,

$$x = \frac{8y+29+10y+25}{-6}$$

$$\Rightarrow x = \frac{18y+54}{-6}$$

$$\Rightarrow -6x = 18y+54$$

$$\boxed{6x+18y+54=0}$$

and

$$x = \frac{8y+29-10y-25}{-6}$$

$$\therefore x = \frac{-2y+4}{-6}$$

$$\therefore -6x = -2y+4$$

$$\boxed{6x-2y+4=0}$$

Here,

$$6x-2y+4=0 \quad \cdots (I)$$

$$6x+18y+54=0 \quad \cdots (II)$$

Formula

$ax+by+c$ এর ঢাল $\boxed{m=-\frac{a}{b}}$

Slope of ①, $m_1=\frac{-6}{-2}=3$, $m_2=-\frac{6}{18}=-\frac{1}{3}$

Since. $m_1 \cdot m_2 = 3\left(-\frac{1}{3}\right)=-1$

Thus the lines are perpendicular.

Class Work

1 * $2y^2-xy-x^2+y+2x-1=0$

H.W 2 * $x^2-5xy-6y^2+14x+5y+4=0$

H.W 3 * $2y^2+3xy-5y-6x+2=0$

} Next class এ জিজ্ঞাসা করবে

Sol(1) | C.W |

$2y^2-xy-x^2+y+2x-1=0$

$\Rightarrow -x^2+x(-y+2)+(2y^2+y-1)=0$

Comparing with the quadratic equation $ax^2+bx+c=0$

here, $a=-1$

$b=-y+2$

$c=2y^2+y-1$

Thus the roots are,

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$\Rightarrow x=\frac{-(-y+2)\pm\sqrt{(-y+2)^2-4(-1)(2y^2+y-1)}}{2(-1)}$$

$$\Rightarrow x = \frac{y-2 \pm \sqrt{y^2-4y+4+8y^2+4y-4}}{-2}$$

$$\Rightarrow x = \frac{y-2 \pm \sqrt{9y^2}}{-2}$$

$$\Rightarrow x = \frac{y-2 \pm 3y}{-2}$$

$$\Rightarrow -2x = y-2 \pm 3y$$

$$\Rightarrow 2x + y \pm 3y - 2 = 0$$

$$\therefore 2x + 4y - 2 = 0 \quad \ldots (i)$$

and $2x - 2y - 2 = 0 \quad \ldots (ii)$

~~Slope of (i), $m = \frac{-2}{4} \Rightarrow -\frac{1}{2}$, $m_2 = \frac{(-2)}{2} = 1$~~

Slope of (i), $m_1 = \frac{-2}{4} = -\frac{1}{2}$, $m_2 = \frac{-2}{-2} = 1$

Thus the lines are neither parallel nor perpendicular.

H.W

2. $x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$

$\Rightarrow x^2 + x(-5y + 14) + (-6y^2 + 5y + 4) = 0$

Comparing with the quadratic equation $ax^2 + bx + c = 0$

here, $a = 1$, $b = (-5y + 14)$ and $c = -6y^2 + 5y + 4$

Thus the roots are,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow x = \frac{-(-5y + 14) \pm \sqrt{(-5y + 14)^2 - 4 \cdot 1 \cdot (-6y^2 + 5y + 4)}}{2 \cdot 1}$$

$$\Rightarrow x = \frac{5y - 14 \pm \sqrt{25y^2 - 140y + 196 + 24y^2 - 20y \pm 16}}{2}$$

$$\Rightarrow 2x = 5y - 14 \pm \sqrt{49y^2 - 160y + 180}$$

$\Rightarrow$

~~Questions & Answers | প্রশ্ন ও উত্তর | প্রশ্ন ও উত্তর | 4 ও 34~~

~~2. $x^2 - 5xy - 6y^2 + 14x + 5y + 34 = 0$~~

~~$\Rightarrow x^2 + x(-5y + 14) + (-6y^2 + 5y + 34) = 0$~~

ধন্যবাদ

H.W

3] $2y^2+3xy-5y-6x+2=0$

$\Rightarrow 2y^2+y(3x-5)+(-6x+2)=0$

Comparing with the quadratic equation $ay^2+by+c=0$ here, $a=2$, $b=3x-5$ and $c=-6x+2$

Thus the roots are,

$$y = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$\Rightarrow y = \frac{-(3x-5) \pm \sqrt{(3x-5)^2-4\times 2\times(-6x+2)}}{2\times 2}$$

$$\Rightarrow 4y = -3x+5 \pm \sqrt{9x^2-30x+25+48x-16}$$

$$\Rightarrow 4y = -3x+5 \pm \sqrt{9x^2+18x+9}$$

$$\Rightarrow 4y = -3x+5 \pm \sqrt{(3x)^2+2\cdot 3x\cdot 3+3^2}$$

$$\Rightarrow 4y = -3x+5 \pm \sqrt{(3x+3)^2}$$

$$\Rightarrow 4y = -3x+5 \pm (3x+3)$$

Now,

$4y = -3x+5+(3x+3)$

$\Rightarrow 4y = 8$

$\Rightarrow 4y-8=0$

$\Rightarrow y-2=0 \quad \cdots (1)$

And,

$4y = -3x+5-(3x+3)$

$\Rightarrow 4y = -3x+5-3x-3$

$\Rightarrow 4y+6x-2=0$

$\Rightarrow 3x+2y-1=0 \quad \cdots (11)$

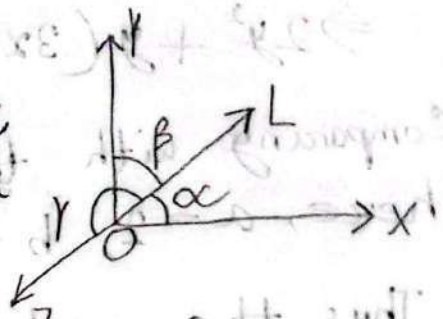
Slope of (1), $m_1 = -\frac{0}{1} = 0$, $m_2 = -\frac{3}{2}$

Thus the lines are neither parallel nor perpendicular.

09 August 2026

Direction Cosine of a line

If a given straight line OL makes angles α, β, γ with x-axis, y-axis, z-axis respectively then $\cos\alpha, \cos\beta, \cos\gamma$ are known as the direction cosine of the straight line OL which is denoted by l, m, n respectively.



That is $l = \cos\alpha,\ m = \cos\beta,\ n = \cos\gamma$

So, the direction cosine of

$$OL = (l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$$

Relation between Direction Cosines:-

The sum of squares of direction cosines of every line is one. That is if l, m, n be the direction cosines of any line, then

$$l^2 + m^2 + n^2 = 1$$

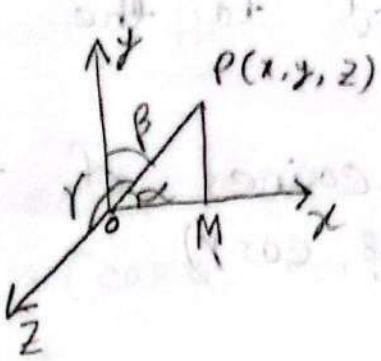
Proof:-

Let the direction cosines of the line OP be $(l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$ as the coordinates of P be (x, y, z)

Here OP is the position vector,

$$OP = \sqrt{x^2 + y^2 + z^2}$$

$\therefore OP^2 = x^2 + y^2 + z^2 \quad \ldots\ldots (i)$



Draw a perpendicular line PM from P on the x-axis then $OM = x$

From $\triangle OPM$, $\cos\alpha = \frac{OM}{OP} = \frac{x}{OP}$

$\therefore x = OP\cos\alpha$

$\Rightarrow x = lOP \quad \ldots\ldots (ii)$

Similarly, $y = m \cdot OP \quad \ldots (iii)$

$z = n \cdot OP \quad \ldots (iv)$

Now, $(ii)^2 + (iii)^2 + (iv)^2$, we get

$x^2 + y^2 + z^2 = l^2 OP^2 + m^2 OP^2 + n^2 OP^2$

$\Rightarrow OP^2 = OP^2(l^2 + m^2 + n^2)$ [Using (i)]

$\Rightarrow l^2 + m^2 + n^2 = 1$

Or, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Define direction cosines and find the relationship between them.

Ex:- Find the direction cosine of the line which is equally inclined to the axes

Soln:-

Let the direction cosines of the line be

$(l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$

Here,

$$\alpha = \beta = \gamma$$

$$\therefore \cos\alpha = \cos\beta = \cos\gamma$$

Thus we can write $l = m = n \quad \ldots (i)$

We know that,

$$l^2 + m^2 + n^2 = 1$$

$$\Rightarrow l^2 + l^2 + l^2 = 1 \quad [\text{using } (i)]$$

$$\Rightarrow 3l^2 = 1$$

$$\therefore l^2 = \frac{1}{3}$$

$$\therefore l = \pm\frac{1}{\sqrt{3}}$$

Similarly,

$$m = \pm\frac{1}{\sqrt{3}}$$

$$n = \pm\frac{1}{\sqrt{3}}$$

So the direction cosines are $(l, m, n) = \left(\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right)$

Ex:- If a line makes angles α, β, γ with the axes, then show that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Sol^n:-

Let the direction cosines of the line be

$(l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$

We know that,

$$l^2 + m^2 + n^2 = 1$$

$$\Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\Rightarrow (1 - \sin^2\alpha) + (1 - \sin^2\beta) + (1 - \sin^2\gamma) = 1$$

$$\Rightarrow 3 - (\sin^2\alpha + \sin^2\beta + \sin^2\gamma) = 1$$

$$\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$$

Direction Ratio

Any three numbers a, b, c which are proportional to the direction cosines l, m, n respectively of a given line are called the direction ratios of the given line.

By definition $a \propto l$

$$\therefore a = kl \quad \ldots (i)$$

k is a constant

$b \propto m$

$$b = km \quad \ldots (ii)$$

- Define direction ratio
- Relationship between direction cosines and direction ratio

$c \propto n$

$c = kn \quad \ldots (iii)$

Now, $(i)^2 + (ii)^2 + (iii)^2$, we get,

$$a^2+b^2+c^2 = l^2k^2 + m^2k^2 + n^2k^2$$

$$\Rightarrow a^2+b^2+c^2 = k^2(l^2+m^2+n^2)$$

$$\Rightarrow a^2+b^2+c^2 = k^2 \cdot 1$$

$$\Rightarrow k^2 = a^2+b^2+c^2$$

$$\Rightarrow k = \pm\sqrt{a^2+b^2+c^2}$$

Putting the values of k in (i), (ii), (iii) we get,

$$a = \pm\sqrt{a^2+b^2+c^2} \times l$$

$$\therefore l = \pm \frac{a}{\sqrt{a^2+b^2+c^2}}$$

Similarly, $m = \pm \frac{b}{\sqrt{a^2+b^2+c^2}}$

and $n = \pm \frac{c}{\sqrt{a^2+b^2+c^2}}$

are the mathematical representation of the relationship between direction cosines and direction ratio of a line.

9 August 2026

Formula 1:-

If a, b, c are direction ratios of the line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ then, $a = x_2 - x_1$, $b = y_2 - y_1$, $c = z_2 - z_1$

Problem:-

If P and Q are $(2, 3, -6)$ and $(3, -4, 5)$ respectively and O be the origin, then find the direction cosines of OP, OQ, PQ.

Soln:-

The coordinate of origin O is $(0, 0, 0)$

Direction Cosines of OP:

Let a_1, b_1, c_1 be the direction ratios of OP and l_1, m_1, n_1 be the direction cosines of OP.

Now, $a_1 = 2 - 0 = 2$

$b_1 = 3 - 0 = 3$

$c_1 = -6 - 0 = -6$

So direction cosines of OP:

$$l_1 = \frac{a_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{2}{\sqrt{2^2 + 3^2 + (-6)^2}} = \frac{2}{\sqrt{49}} = \frac{2}{7}$$

$$m_1 = \frac{b_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{3}{\sqrt{2^2 + 3^2 + (-6)^2}} = \frac{3}{\sqrt{49}} = \frac{3}{7}$$

$$n_1 = \frac{c_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{-6}{\sqrt{2^2 + 3^2 + (-6)^2}} = \frac{-6}{\sqrt{49}} = -\frac{6}{7}$$

OQ = ?

PQ = ?



$l_2 = \frac{3}{5\sqrt{2}}$ $m_2 = \frac{-4}{5\sqrt{2}}$ $n_2 = \frac{1}{\sqrt{2}}$

$l_3 = \frac{1}{\sqrt{171}}$ $m_3 = \frac{-7}{\sqrt{171}}$ $n_3 = \frac{11}{\sqrt{171}}$

Direction Cosines of OQ

Let a_2, b_2, c_2 be the direction ratios of OQ and l_2, m_2, n_2 be the direction cosines of OQ

Now, $a_2 = 3-0 = 3$

$b_2 = -4-0 = -4$

$c_2 = 5-0 = 5$

So direction cosines of OQ:

$$l_2 = \frac{a_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{3}{\sqrt{3^2+(-4)^2+5^2}} = \frac{3}{\sqrt{50}} = \frac{3}{5\sqrt{2}}$$

$$m_2 = \frac{b_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{-4}{\sqrt{3^2+(-4)^2+5^2}} = \frac{-4}{\sqrt{50}} = \frac{-4}{5\sqrt{2}}$$

$$n_2 = \frac{c_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{5}{\sqrt{3^2+(-4)^2+5^2}} = \frac{5}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Direction cosines of PQ

Let a_3, b_3, c_3 be the direction ratios of PQ and l_3, m_3, n_3 be the direction cosines of PQ

Now, $a_3 = 3-2 = 1$

$b_3 = -4-3 = -7$

$c_3 = 5+6 = 11$



So direction cosines of PQ

$$l_3 = \frac{a_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{1}{\sqrt{1^2+(-7)^2+(11)^2}} = \frac{1}{\sqrt{171}}$$

$$m_3 = \frac{b_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{-7}{\sqrt{1^2+(-7)^2+(11)^2}} = \frac{-7}{\sqrt{171}}$$

$$n_3 = \frac{c_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{11}{\sqrt{1^2+(-7)^2+(11)^2}} = \frac{11}{\sqrt{171}}$$

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H.W: If P and Q are (2,3,4) and (1,1,−1) respectively and O be the origin, then find the direction cosines of OP, OQ and PQ

ans: $OP\left(\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}}\right)$ $OQ\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$ $PQ\left(-\frac{1}{\sqrt{30}}, \frac{-2}{\sqrt{30}}, \frac{-5}{\sqrt{30}}\right)$

Formula 2:-

If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines and θ be the angle between them, then

$$\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

Formula 3:-

Condition of perpendicularity of two lines:
If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines and if these two lines are perpendicular, then

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

Formula 4:-

Condition of Parallelism of two lines:
If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines and if these two lines are parallel, then

$$l_1 = l_2, \quad m_1 = m_2, \quad n_1 = n_2$$



Problem:-

Find the cosine of the angle between two lines having the direction cosines $\left(\frac{4}{9}, -\frac{8}{9}, \frac{-1}{9}\right)$ and $\left(-\frac{1}{3}, \frac{-2}{3}, -\frac{2}{3}\right)$

Soln:-

$l_1 = \frac{4}{9}$,

$m_1 = \frac{-8}{9}$,

$n_1 = -\frac{1}{9}$

$l_2 = -\frac{1}{3}$;

$m_2 = \frac{-2}{3}$,

$n_2 = \frac{-2}{3}$



Let θ be the angle between two lines,

then $\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$

$= \left(\frac{4}{9}\right)\cdot\left(-\frac{1}{3}\right) + \left(-\frac{8}{9}\right)\cdot\left(-\frac{2}{3}\right) + \left(-\frac{1}{9}\right)\cdot\left(\frac{-2}{3}\right)$

$= -\frac{4}{27} + \frac{16}{27} + \frac{2}{27}$

$\therefore \cos\theta = \frac{14}{27}$

H.W → Find the cosine of the angle between two lines having the direction cosines $\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and $\left(\frac{1}{2}, \frac{-\sqrt{6}}{\sqrt{4}}, -\frac{1}{2}\right)$

Ans: $\cos\theta = \frac{1}{\sqrt{2}}$

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H.W → Check that joining points from A(5, 2, -3) to Q(6, 1, 4) to line joining the points from C(-3, 2, -1) to L(-1, -4, 1) is parallel or not

Problem:-

Prove that, joining the points from P(2, 3, -2) to Q(3, 1, 1) is parallel to the line joining the points from R(2, 1, -5) to L(4, -3, 1)

Sol^n:-

Direction cosines of PQ:

Let a_1, b_1, c_1 be the direction ratios of PQ and l_1, m_1, n_1 be the direction cosines of PQ

Now,

$a_1 = 3-2 = 1$, $b_1 = 1-3 = -2$ $c_1 = 1-(-2) = 3$

So the direction cosine of PQ:

$$l_1 = \frac{1}{\sqrt{1^2+(-2)^2+3^2}} = \frac{1}{\sqrt{14}}$$

$$m_1 = \frac{-2}{\sqrt{1^2+(-2)^2+3^2}} = \frac{-2}{\sqrt{14}}$$

$$n_1 = \frac{3}{\sqrt{1^2+(-2)^2+3^2}} = \frac{3}{\sqrt{14}}$$



Direction cosines of RL:

Let, a_2, b_2, c_2 be the direction ratios of RL and l_2, m_2, n_2 be the direction cosines of RL

Now, $a_2 = 4-2 = 2$, $b_2 = -3-1 = -4$, $c_2 = 1-(-5) = 6$

So direction cosines of RL:

$$l_2 = \frac{2}{\sqrt{2^2+(-4)^2+6^2}} = \frac{2}{\sqrt{56}} = \frac{2}{2\sqrt{14}} = \frac{1}{\sqrt{14}}$$

$$m_2 = \frac{-4}{\sqrt{2^2+(-4)^2+6^2}} = \frac{-4}{\sqrt{56}} = \frac{-4}{2\sqrt{14}} = \frac{-2}{\sqrt{14}}$$

$$n_2 = \frac{6}{\sqrt{2^2+(-4)^2+6^2}} = \frac{6}{\sqrt{56}} = \frac{6}{2\sqrt{14}} = \frac{3}{\sqrt{14}}$$

Since, $l_1 = l_2$, $m_1 = m_2$ and $n_1 = n_2$, so PQ and RL lines are parallel

Problem: Prove that, the lines AB and BC are perpendicular to each other when A(-11, 8, 4) B(-1, -7, -1), C(9, -2, 4)

Soln:

D.C of AB:

Let a_1, b_1, c_1 be the direction ratios of AB and l_1, m_1, n_1 be the direction cosines of AB

$a_1 = -1 - (-11) = 10$

$b_1 = -7 - 8 = -15$

$c_1 = -1 - 4 = -5$

So direction cosines of AB:

$$l_1 = \frac{10}{\sqrt{10^2+(-15)^2+(-5)^2}} = \frac{10}{\sqrt{350}}$$

$$m_1 = \frac{-15}{\sqrt{10^2+(-15)^2+(-5)^2}} = \frac{-15}{\sqrt{350}}$$

$$n_1 = \frac{-5}{\sqrt{10^2+(-15)^2+(-5)^2}} = \frac{-5}{\sqrt{350}}$$



D.C. of BC:-

Let a_2, b_2, c_2 be the direction ratios of BC and l_2, m_2, n_2 be the direction cosines of BC

Now,

$a_2 = 9 - (-1) = 10$,

$b_2 = -2 - (-7) = 5$

$c_2 = 4 - (-1) = 5$

So DC of BC:

$$l_2 = \frac{10}{\sqrt{10^2+5^2+5^2}} = \frac{10}{\sqrt{150}}$$

$$m_2 = \frac{5}{\sqrt{10^2+5^2+5^2}} = \frac{5}{\sqrt{150}}$$

$$n_2 = \frac{5}{\sqrt{10^2+5^2+5^2}} = \frac{5}{\sqrt{150}}$$

Now, $l_1l_2 + m_1m_2 + n_1n_2$

$= \frac{10}{\sqrt{350}} \times \frac{10}{\sqrt{150}} + \left(\frac{-15}{\sqrt{350}} \times \frac{5}{\sqrt{150}}\right) + \left(\frac{-5}{\sqrt{350}} \times \frac{5}{\sqrt{150}}\right)$

$= 0$ ∴ The lines are perpendicular

H:W

Prove that the lines AB and CD are perpendicular to each other where.

A(6, -6, 0),

B(-1, -7, 6), C(3, -4, 4)

D(2, -9, 2)

H.W.

1] If P and Q are (2, 3, 4) and (1, 1, −1) respectively and O be the origin, then find the direction Cosines of OP, OQ and PQ.

Sol^n:

The coordinate of origin O is (0, 0, 0)

Direction cosines of OP:

Let a_1, b_1, c_1 be the direction ratios of OP and l_1, m_1, n_1 be the direction cosines of OP

Now,

$a_1 = 2-0 = 2$, $b_1 = 3-0 = 3$, $c_1 = 4-0 = 4$

So direction cosines of OP:

$$l_1 = \frac{a_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{2}{\sqrt{2^2+3^2+4^2}} = \frac{2}{\sqrt{29}}$$

$$m_1 = \frac{a_2}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{3}{\sqrt{2^2+3^2+4^2}} = \frac{3}{\sqrt{29}}$$

$$n_1 = \frac{a_3}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{4}{\sqrt{2^2+3^2+4^2}} = \frac{4}{\sqrt{29}}$$

Direction cosines of OQ:

Let a_2, b_2, c_2 be the direction ratios of OQ and l_2, m_2, n_2 be the direction cosines of OQ

Now, $a_2 = 1-0 = 1$, $b_2 = 1-0 = 1$, $c_2 = -1-0 = -1$

So direction cosines of OQ:

$$l_2 = \frac{a_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{1}{\sqrt{1^2+1^2+(-1)^2}} = \frac{1}{\sqrt{3}}$$

$$m_2 = \frac{b_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{1}{\sqrt{1^2+1^2+(-1)^2}} = \frac{1}{\sqrt{3}}$$

$$n_2 = \frac{c_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{-1}{\sqrt{1^2+1^2+(-1)^2}} = \frac{-1}{\sqrt{3}}$$



H.W

Direction cosines of PQ:

Let a_3, b_3, c_3 be the direction ratios of PQ and l_3, m_3, n_3 be the direction cosines of PQ

Now,

$a_3 = 1-2 = -1$, $b_3 = 1-3 = -2$, $c_3 = -1-4 = -5$

So direction cosines of PQ:

$$l_3 = \frac{a_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{-1}{\sqrt{(-1)^2+(-2)^2+(-5)^2}} = \frac{-1}{\sqrt{1+4+25}} = \frac{-1}{\sqrt{30}}$$

$$m_3 = \frac{b_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{-2}{\sqrt{(-1)^2+(-2)^2+(-5)^2}} = \frac{-2}{\sqrt{1+4+25}} = \frac{-2}{\sqrt{30}}$$

$$n_3 = \frac{c_3}{\sqrt{a_3^2+b_3^2+c_3^2}} = \frac{-5}{\sqrt{(-1)^2+(-2)^2+(-5)^2}} = \frac{-5}{\sqrt{1+4+25}} = \frac{-5}{\sqrt{30}}$$

2| Find the cosine of the angle between two lines having the direction cosines $\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and $\left(\frac{1}{2}, \sqrt{\frac{6}{4}}, -\frac{1}{2}\right)$

Soln:

$l_1 = \frac{1}{\sqrt{3}}$, $m_1 = \frac{-1}{\sqrt{3}}$, $n_1 = \frac{1}{\sqrt{3}}$

$l_2 = \frac{1}{2}$, $m_2 = \sqrt{\frac{6}{4}}$, $n_2 = \frac{-1}{2}$

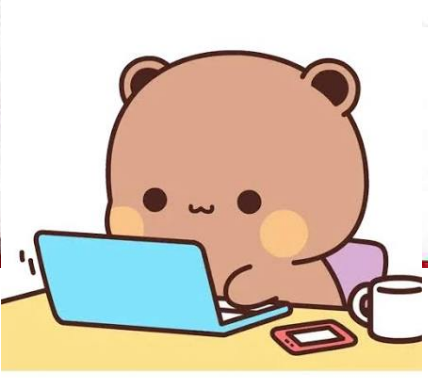
Let θ be the angle between two lines, then, $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$

$$= \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{3}}\right) + \left(-\frac{1}{\sqrt{3}}\right)\left(\sqrt{\frac{6}{4}}\right) + \left(\frac{1}{\sqrt{3}}\right)\left(\frac{-1}{2}\right)$$

$$= \frac{1}{2\sqrt{3}} + \frac{\sqrt{6}}{2\sqrt{3}} - \frac{1}{2\sqrt{3}}$$

$$= \frac{1+\sqrt{6}-1}{2\sqrt{3}}$$

$$= \frac{1}{\sqrt{2}}$$



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H.W.



3] Check that joining points from A(5,2,-3) to Q(6,1,4) to line joining the points from C(-3,2,-1) to L(-1,-4,13) is parallal or not

Soln:-

Direction cosines of AQ:

Let a_1, b_1, c_1 be the direction ratios of AQ and l_1, m_1, n_1 be the direction cosines of AQ

now, $a_1 = 6-5 = 1$, $b_1 = 1-2 = -1$, $c_1 = 4+3 = 7$

So, the direction cosines of AQ

$$l_1 = \frac{a_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{1}{\sqrt{(1)^2+(-1)^2+7^2}} = \frac{1}{\sqrt{1+1+49}} = \frac{1}{\sqrt{51}}$$

$$m_1 = \frac{b_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{-1}{\sqrt{(1)^2+(-1)^2+7^2}} = \frac{-1}{\sqrt{1+1+49}} = \frac{-1}{\sqrt{51}}$$

$$n_1 = \frac{c_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{7}{\sqrt{(1)^2+(-1)^2+(7)^2}} = \frac{7}{\sqrt{1+1+49}} = \frac{7}{\sqrt{51}}$$

Direction cosines of CL:

Let a_2, b_2, c_2 be the direction ratios of CL and l_2, m_2, n_2 be the direction cosines of CL

now, $a_2 = -1+3 = 2$, $b_2 = -4-2 = -6$, $c_2 = 13+1 = 14$

So, the direction cosines of CL

$$l_2 = \frac{a_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{2}{\sqrt{2^2+(-6)^2+(14)^2}} = \frac{2}{\sqrt{236}} = \frac{2}{2\sqrt{59}} = \frac{1}{\sqrt{59}}$$

$$m_2 = \frac{b_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{-6}{\sqrt{2^2+(-6)^2+(14)^2}} = \frac{-6}{2\sqrt{59}} = \frac{-3}{\sqrt{59}}$$

$$n_2 = \frac{c_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{14}{2\sqrt{59}} = \frac{7}{\sqrt{59}}$$

Since, $l_1 \neq l_2$, $m_1 \neq m_2$, $n_1 \neq n_2$

So, AQ and CL lines are not parallal

H.W

4] Prove that the lines AB and CD are perpendicular to each other where A(6,-6,0), B(-1,-7,6), C(3,-4,4), D(2,-9,2)

Sol^n:-

Direction cosines of AB:

Let a_1, b_1, c_1 be the direction ratios of AB and l_1, m_1, n_1 be the direction cosines of AB

$a_1 = -1-6 = -7$, $b_1 = -7+6 = -1$, $c_1 = 6-0 = 6$

So direction cosines of AB:

$$l_1 = \frac{a_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{-7}{\sqrt{(-7)^2+(-1)^2+6^2}} = \frac{-7}{\sqrt{49+1+36}} = \frac{-7}{\sqrt{86}}$$

$$m_1 = \frac{a_2}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{-1}{\sqrt{(-7)^2+(-1)^2+6^2}} = \frac{-1}{\sqrt{49+1+36}} = \frac{-1}{\sqrt{86}}$$

$$n_1 = \frac{a_3}{\sqrt{a_1^2+b_1^2+c_1^2}} = \frac{6}{\sqrt{(-7)^2+(-1)^2+6^2}} = \frac{6}{\sqrt{49+1+36}} = \frac{6}{\sqrt{86}}$$

Direction cosines of CD:

Let a_2, b_2, c_2 be the direction ratios of CD and l_2, m_2, n_2 be the direction cosines of CD

$a_2 = 2-3 = -1$, $b_2 = -9+4 = -5$, $c_2 = 2-4 = -2$

So direction cosines of CD:

$$l_2 = \frac{a_2}{\sqrt{a_2^2+b_2^2+c_2^2}} = \frac{-1}{\sqrt{(-1)^2+(-5)^2+(-2)^2}} = \frac{-1}{\sqrt{1+25+4}} = \frac{-1}{\sqrt{30}}$$

$$m_2 = \frac{-5}{\sqrt{30}}$$

$$n_2 = \frac{-2}{\sqrt{30}}$$

Now, $l_1 l_2 + m_1 m_2 + n_1 n_2$

$$= \left(\frac{-7}{\sqrt{86}}\right)\left(\frac{-1}{\sqrt{30}}\right) + \left(\frac{-1}{\sqrt{86}}\right)\left(\frac{-5}{\sqrt{30}}\right) + \left(\frac{6}{\sqrt{86}}\right)\left(\frac{-2}{\sqrt{30}}\right)$$

$$= 0$$

∴ The lines are perpendicular

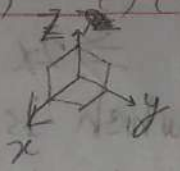


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18/8/2026

Plane



General Equation of Plane:

$ax+by+cz+d=0$; where a, b, c, d are constants.

Formula 1: General equation of plane passing through the origin $(0,0,0)$ is $ax+by+cz=0$

Formula 2: Equation of plane passing through the point (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) is

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

Formula 3: The perpendicular distance from the point (x_1, y_1, z_1) to the plane $ax+by+cz=0$ is

$$D = \left| \frac{ax_1+by_1+cz_1+d}{\sqrt{a^2+b^2+c^2}} \right|$$

Find equation of plane passing through $(2,3,1)$, $(1,1,3)$, $(2,2,3)$. Also find perpendicular distance from the point $(5,6,7)$

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 2 & 2 & 3 & 1 \end{vmatrix} = \begin{vmatrix} x-2 & y-3 & z-1 & 0 \\ 1 & 2 & -2 & 0 \\ -1 & -1 & 0 & 0 \\ 2 & 2 & 3 & 1 \end{vmatrix} \quad \begin{bmatrix} R_1' = R_1 - R_2 \\ R_2' = R_2 - R_3 \\ R_3' = R_3 - R_4 \end{bmatrix}$$

$$= -1\{(y-3)(-2) - 2(z-1)\} + 1\{(x-2)(-2) - (z-1)\}$$

$$= 2y - 6 + 2z - 2 - 2x + 4 - z + 1 = -2x + 2y + z - 3 = 0$$

$\Rightarrow 2x-2y-z+3=0$

which is the required equation of a plane

The perpendicular distance from the point (5,6,7) to this plane is

~~Practice: Fin~~ $\left|\frac{2\times5-2\times6-1\times7+3}{\sqrt{2^2+(-2)^2+(-1)^2}}\right|$

$$=\left|\frac{10-12-7+3}{\sqrt{4+4+1}}\right|=\left|\frac{-6}{\sqrt{9}}\right|=|-2|=2$$

Practiee:

Find the equation of plane which is passing through (2,1,3), (-1,-2,4), (4,2,1). Also find the perpendicular distance from the point (1, 1, -1) to this plane.

$\Rightarrow$ The equation of plane passing through (2, 1, 3), (-1, -2, 4), (4, 2, 1) is

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 1 & 3 & 1 \\ -1 & -2 & 4 & 1 \\ 4 & 2 & 1 & 1 \end{vmatrix} = 0$$

here,

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 1 & 3 & 1 \\ -1 & -2 & 4 & 1 \\ 4 & 2 & 1 & 1 \end{vmatrix} = \begin{vmatrix} x & y & z & 1 \\ 2-x & 1-y & 3-z & 0 \\ -3 & -3 & 1 & 0 \\ 5 & 4 & -3 & 0 \end{vmatrix}$$

$[R_4' = R_4 - R_3$
$R_3' = R_3 - R_2$
$R_2' = R_2 - R_1]$

$$= 1\begin{vmatrix} 2-x & 1-y & 3-z \\ -3 & -3 & 1 \\ 5 & 4 & -3 \end{vmatrix}$$

$$= 5\{(1-y)\times 1 - (-3)(3-z)\} - 4\{(2-x)\times 1 - (-3)(3-z)\} - 3\{(2-x)(-3) - (-3)(1-y)\}$$

$$= 5(1-y+9-3z) - 4(2-x+9-3z) - 3(-6+3x+3-3y)$$

$$= 5-5y+45-15z-8+4x-36+12z+18-9x-9+9y$$

$$= -5x+4y-3z+15=0$$

∴ Required equation

$$-5x+4y-3z+15=0$$

$$\Rightarrow 5x-4y+3z-15=0$$

which is the required equation of a plane

The perpendicular distance from the point $(1,1,-1)$ to this plane is

$$\left|\frac{5\times 1-4\times 1+3\times(-1)-15}{\sqrt{5^2+(-4)^2+3^2}}\right|$$

$$= \left|\frac{5-4-3-15}{\sqrt{50}}\right|$$

$$= \left|\frac{-17}{\sqrt{50}}\right|$$

$$= \frac{17}{\sqrt{50}}$$

Formula 4:

The four point $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3), (x_4, y_4, z_4)$ are coplanar if their determinant is zero.

Problem:
Show that the four points $(0, -1, -1), (4, 5, 1), (3, 9, 4), (-4, 4, 4)$ are coplanar.

Solution:

To prove that the four points are coplanar, we need to show their determinant is zero,

Now,
$$\begin{vmatrix} 0 & -1 & -1 & 1 \\ 4 & 5 & 1 & 1 \\ 3 & 9 & 4 & 1 \\ -4 & 4 & 4 & 1 \end{vmatrix}$$

$$= 0\begin{vmatrix} 5 & 1 & 1 \\ 9 & 4 & 1 \\ 4 & 4 & 1 \end{vmatrix} + 1\begin{vmatrix} 4 & -1 & 1 \\ 3 & 4 & 1 \\ -4 & 4 & 1 \end{vmatrix} - 1\begin{vmatrix} 4 & 5 & 1 \\ 3 & 9 & 1 \\ -4 & 4 & 1 \end{vmatrix} - 1\begin{vmatrix} 4 & 5 & 1 \\ 3 & 9 & 4 \\ -4 & 4 & 4 \end{vmatrix}$$

$$= 0 + 1\{4(4-4) - 1(3+4) + 1(12+16)\} - 1\{4(9-4) - 5(3+4) + 1(12+36)\} - 1\{4(36-16) - 5(12+16) + 1(12+36)\}$$

$$= (-7+28) - (20 - 35 + 48) - (80 - 140 + 48)$$

$$= -7 + 28 - 20 + 35 - 48 - 80 + 140 - 48 = 0$$

So these four points are coplanar.

[Proved]

Practice

Show that the four points $(-6, 3, 2)$, $(3, -2, 4)$, $(5, 7, 3)$, $(-13, 17, -1)$ are coplanar

Solution

To prove that the four points are coplanar, we need to show their determinant is zero,

Now, $\begin{vmatrix} -6 & 3 & 2 & 1 \\ 3 & -2 & 4 & 1 \\ 5 & 7 & 3 & 1 \\ -13 & 17 & -1 & 1 \end{vmatrix}$

$= -6\begin{vmatrix} -2 & 4 & 1 \\ 7 & 3 & 1 \\ 17 & -1 & 1 \end{vmatrix} - 3\begin{vmatrix} 3 & 4 & 1 \\ 5 & 3 & 1 \\ -13 & -1 & 1 \end{vmatrix} + 2\begin{vmatrix} 3 & -2 & 1 \\ 5 & 7 & 1 \\ -13 & 17 & 1 \end{vmatrix} - 1\begin{vmatrix} 3 & -2 & 4 \\ 5 & 7 & 3 \\ -13 & 17 & -1 \end{vmatrix}$

$= -6\{(-2)(3+1) - 4(7-17) + 1(-7-51)\}$
$\quad -3\{3(3+1) - 4(5+13) + 1(-5+39)\}$
$\quad +2\{3(7-17) + 2(5+13) + 1(25+91)\}$
$\quad -1\{3(-7-51) + 2(-5+39) + 4(35+91)\}$

$= -6\{-8 + 40 - 58\} - 3\{12 - 72 + 34\} + 2\{-30 + 36 + 176\}$
$\quad -1\{-174 + 68 + 704\}$

~~$= 48 - 240 + 348 - 36 + 216 - 102 + 472 + 172 + 174 - 68 - 704$~~
~~-60~~ ~~352~~

$= -6(-26) - 3(-26) + 2 \times 182 - 1 \times 598$

$= 0$

So the four points are coplanar

[Proved]

23/08/2026

Formula 5

Angle Between two planes:

If $a_1x + b_1y + c_1z + d_1 = 0$
and $a_2x + b_2y + c_2z + d_2 = 0$
are two planes, and θ be the angle between them,

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$



Problem:

Find the angle between two planes $2x - 2y - z + 3 = 0$ and $5x - 4y + 3z - 15 = 0$.

Soln:-

Let θ be the angle between these two planes,

$$\cos\theta = \frac{2\cdot5 + (-2)\times(-4) + (-1)\times3}{\sqrt{\{2^2+(-2)^2+(-1)^2\}\{5^2+(-4)^2+3^2\}}}$$

$$\Rightarrow \cos\theta = \frac{15}{\sqrt{9\times50}} = \frac{15}{3\times5\sqrt{2}}$$

$$\Rightarrow \cos\theta = \frac{1}{\sqrt{2}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \cos^{-1}\left(\cos\frac{\pi}{4}\right)$$

$$\therefore \theta = \frac{\pi}{4}$$

H.W

Find the angle between two planes

$2x - y + z - 6 = 0$
and $x + y + 2z - 7 = 0$

Ans: $\theta = \frac{\pi}{3}$

Formula 6:-

Condition of perpendicularity of two planes:

If $a_1x + b_1y + c_1z + d_1 = 0$ and
$a_2x + b_2y + c_2z + d_2 = 0$ are two ~~perpendi~~ planes, and they are perpendicular to each other, then

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

Formula 7:

Condition of paralelism of two planes:

If $a_1x + b_1y + c_1z + d_1 = 0$ and
$a_2x + b_2y + c_2z + d_2 = 0$ are two plane, and they are paralled to each other, then

$$a_1 = a_2 ; \; b_1 = b_2 ; \; c_1 = c_2$$

Formula 8:

The equation of plane parallel to a given plane $ax + by + cz + d = 0$ is $ax + by + cz + k = 0$



Problem:-

Find the equation of plane which is passing through $(1,2,2)$ and parallel to $3x+2y+z-7=0$.

Soln:-

The equation of plane parallel to $3x+2y+z-7=0$ is $3x+2y+z+k=0 \cdots\cdots (1)$

As, (1) is passing through $(1,2,2)$, thus

$$(3\times1)+(2\times2)+2+k=0$$
$$\Rightarrow 9+k=0$$
$$\therefore k=-9$$

Putting the value of k in (1) we get,

$$3x+2y+z-9=0$$

(*) Find the equation of plane which is passing through $(4,0,1)$ and parallel to $4x+3y-12z+6=0$

Soln:-

The equation of plane parallel to $4x+3y-12z+6=0$ is $4x+3y-12z+k=0 \cdots (1)$

As (1) is passing through $(4,0,1)$, thus

$$4\times4+3\times0-12\times1+k=0$$
$$\Rightarrow 4+k=0$$
$$\therefore k=-4$$

Putting the value of k in (1) we get,

$$4x+3y-12z-4=0$$

Ans: $x+3y+z-2=0$

H.W.1 Find the equation plane which is passing through $(1,1,2)$ and perpendicular to the planes $2x-2y-4z-6=0$ and $3x+y+6z-4=0$

Formula 9:-

Equation of plane passing through the point (x_1, y_1, z_1) is $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$

Problem:-

Find the equation of plane which is passing through $(-1,3,2)$ and perpendicular to two planes $x+2y+2z-5=0$ and $3x+3y+2z-8=0$

Soln:-

The equation of plane passing through $(-1,3,2)$ is $a(x+1)+b(y-3)+c(z-2)=0 \ldots\ldots (I)$

Since, (I) is perpendicular to two planes

$x+2y+2z-5=0$ and $3x+3y+2z-8=0$

So we get,

$1.a+2.b+2c=0 \ldots\ldots (II)$

$3.a+3b+2c=0 \ldots\ldots (III)$

By cross multiplication is (II), (III) we get,

$$\frac{a}{4-6}=\frac{-b}{2-6}=\frac{c}{3-6}$$

$$\therefore \frac{a}{-2}=\frac{b}{4}=\frac{c}{-3}=k \text{ (say)}$$

So, $a=-2k$, $b=4k$, $c=-3k$

Putting the values of a, b, c in (I) we get,

$-2k(x+1)+4k(y-3)-3k(z-2)=0$

$\Rightarrow -2(x+1)+4(y-3)-3(z-2)=0$

$\Rightarrow -2x-2+4y-12-3z+6=0$

$\Rightarrow -2x+4y-3z-8=0$

Which is the required plane.

H.W

Find the equation of plane which is passing through $(1,-1,1)$ and the intersection of two planes $x-2y+3z+4=0$ and $2x-3y+4z-7=0$ [Ans: $9x-5y+17z-39=0$]

Formula 10:-

The equation of plane passing through the interaction of two planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ is

$$(a_1x+b_1y+c_1z+d_1)+K(a_2x+b_2y+c_2z+d_2)=0$$

Problem:-

Find the equation of plane which is passing through $(1,2,3)$ and the intersection of two planes $x+2y+3z+4=0$ and $4x+3y+2z+1=0$

Solution:-

The equation of plane passing through the intersection of two planes is

$$(x+2y+3y+4)+K(4x+3y+2z+1)=0 \quad \cdots\cdots (1)$$

we get,

$$1+(2\times 2)+3\times 3+4+K\{(4\times 1)+(3\times 2)+(2\times 3)+1\}=0$$

$$\Rightarrow (1+4+9+4)+K(4+6+6+1)=0$$

$$\Rightarrow 18+17K=0$$

$$\therefore K=-\frac{18}{17}$$

Putting the value of K in (1) we get,

$$(x+2y+3z+4)-\frac{18}{17}(4x+3y+2z+1)=0$$

$$\Rightarrow 17(x+2y+3z+4)-18(4x+3y+2z+1)=0$$

$$\Rightarrow 17x+34y+51z+68-72x-54y-36z-18=0$$

$$\Rightarrow -11x-4y+3z+10=0$$

which is the required plane

Problem:

Find the equation of plane which is parallel to $4x - 4y + 7z - 3 = 0$ and a distance 4 unit from point $(4, 1, -2)$

Sol^n:-

The equation of plane parallel to $4x - 4y + 7z - 3 = 0$ is

$$4x - 4y + 7z + K = 0 \quad \cdots \textcircled{1}$$

The distance from point $(4, 1, -2)$ to ① is

$$\pm \frac{4\times4 - 4\times1 + 7\times(-2) + K}{\sqrt{4^2 + (-4)^2 + (7)^2}}$$

Thus $\pm\left(\frac{-2+K}{\sqrt{81}}\right) = 4$

$\Rightarrow \pm(-2+K) = 36$



Now,

$-2 + K = 36$

$\therefore K = 38$

$-(-2+K) = 36$

$\Rightarrow 2 - K = 36$

$\therefore K = -34$

Putting the value of K in ① we get

$$\begin{cases} 4x - 4y + 7z + 38 = 0 \\ 4x - 4y + 7z - 34 = 0 \end{cases}$$

H.W

Find the equation of plane which is parallel to plane $2x - 3y - 6z - 14 = 0$ and a distance 5 unit from the origin.

Ans: $\begin{cases} 2x - 3y - 6z + 35 = 0 \\ 2x - 3y - 6z - 35 = 0 \end{cases}$

CamScanner

H.W. Starts here

1. Find the angle between two planes $2x-y+z-6=0$ and $x+y+2z-7=0$

Ans:

Let θ be the angle between these two planes,

$$\cos\theta = \frac{2\times1+(-1)\times1+(1)\times2}{\sqrt{2^2+(-1)^2+(1)^2}\sqrt{1^2+1^2+2^2}}$$

$$\Rightarrow \cos\theta = \frac{2-1+2}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

2. Find the equation plane which is passing through $(1,1,2)$ and perpendicular to the plane $2x-2y-4z-6=0$ and $3x+y+6z-4=0$

Ans:

The equation of plane passing through $(1,1,2)$ is

$$a(x-1)+b(y-1)+c(z-2)=0 \quad \ldots (1)$$

Since (1) is perpendicular to two planes $2x-2y-4z-6=0$ and $3x+y+6z-4=0$

So we get,

$$2a-2b-4c=0 \quad \ldots (2)$$

$$3a+b+6c=0 \quad \ldots (3)$$

By cross multiplication in (2) and (3) we get-

$$\frac{a}{-12+4} = \frac{b}{-12-12} = \frac{c}{2+6}$$

$$\Rightarrow \frac{a}{-8} = \frac{b}{-24} = \frac{c}{8}$$

$$\Rightarrow \frac{a}{-1} = \frac{b}{-3} = c = K \text{ (say)}$$

So, $a=-K$, $b=3K$, $c=K$

Cross multiplication

$a_1x+b_1y+c_1z=0$
$a_2x+b_2y+c_2z=0$

$a_1 \; b_1 \; c_1 \; a_1$
$a_2 \; b_2 \; c_2 \; a_2$

$\Rightarrow \frac{x}{b_1c_2-b_2c_1} = \frac{y}{c_1a_2-c_2a_1} = \frac{z}{a_1b_2-a_2b_1}$

Putting the values of a, b, c in (1) we get,

$-1K(x-1)-3K(y-1)+1K(z-2)=0$

$\Rightarrow -1(x-1)-3(y-1)+1(z-2)=0$

$\Rightarrow -1x+1-3y+3+z-2=0$

$\Rightarrow x+3y-z-2=0$

3. Find the equation of plane which is passing through $(1,-1,1)$ and the intersection of two planes $x-2y+3z+4=0$ and $2x-3y+4z-7=0$

Ans:

The equation of plane passing through the intersection of two planes is

$$x-2y+3z+4+K(2x-3y+4z-7)=0 \quad \ldots (1)$$

Since (1) is passing through $(1,-1,1)$, we we get,

$1-2\times(-1)+3\times1+4+K(2\times1-3\times(-1)+4\times1-7)=0$

$\Rightarrow 1+2+3+4+K(2+3+4-7)=0$

$\Rightarrow 10+K(2)=0$

$\Rightarrow K=-\frac{10}{2}=-5$

Putting value of K in (1) we get

$x-2y+3z+4-5(2x-3y+4z-7)=0$

$\Rightarrow x-2y+3z+4-10x+15y-20z+35=0$

$\Rightarrow -9x+13y-17z+39=0$

$\Rightarrow 9x-13y+17z-39=0$ which is the required plane

A.W ends here

4. Find the equation of plane which is parallel to plane $2x-3y-6z-14=0$ and a distance 5 unit from the origin

Ans:

The equation of plane parallel $2x-3y-6z-14=0$ is $2x-3y-6z+K=0$ —(1)

The distance from origin $(0,0,0)$ is ~~5 unit~~

$$\pm \frac{2\times0-3\times0-6\times0+K}{\sqrt{2^2+(-3)^2+(-6)^2}}$$



Thus $\pm \frac{K}{\sqrt{4+9+36}} = 5$

$\Rightarrow \pm \frac{K}{\sqrt{49}} = 5$

$\Rightarrow \pm \frac{K}{7} = 5$

$\Rightarrow K = \pm 35$

$\therefore K = 35$ and $K = -35$

Putting the value of K in (1) we get

$2x-3y-6z+35=0$

$2x-3y-6z-35=0$

25.08.2026

Defination exam এ আসবে পাবে

upper-triangular matrix

Echelon matrix (row echelon form)

① All zero rows appear at the bottom.

② Each non-zero row has a leading entry (Pivot), and each pivot is strictly to the right of the pivot in the row above.

③ All entries below each pivot are zero

$$\begin{pmatrix} 3 & 2 & 0 & 7 \\ 0 & 4 & 5 & 10 \\ 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 3 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

Reduced row echelon matrix

* উপরের ①, ②, ③ Copy-Paste

④ Each pivot equals 1.

⑤ Each pivot is the only non-zero entry in its column

$$\begin{pmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

* ● → Any scaler $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

☆ Define Echelon matrix
☆ Define row echelon matrix

Mid এ আসবে ☆☆☆☆☆

দুটি Matrix সমান হলে সবগুলোর অবস্থান একিরকম থাকে। নিচে তাই (=) Use না করে (~) করা হয়েছে।

Ex:-
Find the Echelon form of $\begin{pmatrix} 3 & 4 & -1 & 2 \\ 1 & -2 & 3 & 1 \\ 3 & 14 & -11 & 1 \end{pmatrix}$

Sol^n:
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 3 & 4 & -1 & 2 \\ 3 & 14 & 11 & 1 \end{pmatrix} [R_1 \longleftrightarrow R_2]$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 10 & -10 & -1 \\ 0 & 20 & -20 & -2 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 10 & -10 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} [R_3' = R_3 - 2R_2]$$

Ex:-
Find the echelon form of $\begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 4 \\ 0 & 7 & 10 \end{pmatrix}$

Sol^n
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 10 \\ 0 & 7 & 10 \end{pmatrix} [R_2' = R_2 + 2R_1]$$

$$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 10 \\ 0 & 0 & 0 \end{pmatrix} [R_3' = R_3 - R_2]$$

Ex :- Find the echelon form of $\begin{pmatrix} 1 & -1 & 2 & 0 \\ 1 & 1 & 1 & 1 \\ -1 & -3 & 0 & -2 \end{pmatrix}$

Sol^n

Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & -4 & 2 & 2 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 + R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{bmatrix} R_3' = R_3 + 2R_2 \end{bmatrix}$$

H.W:

(1) $\begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$ (2) $\begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & 4 \end{pmatrix}$

(3) $\begin{pmatrix} 2 & 1 & -1 & 2 \\ -1 & 3 & 2 & 3 \\ 1 & 2 & 1 & -1 \\ 2 & -3 & -1 & 4 \end{pmatrix}$

H.W.

① Find the echelon form of $\begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$

Sol^n
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - R_2 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} R_3' = R_3 + R_2 \end{bmatrix}$$

② Find the echelon form of $\begin{pmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & 4 \end{pmatrix}$

Sol^n
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 1 & 1 & 3 \\ 0 & 2 & -6 \\ 0 & 2 & -2 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 + 2 \times R_2 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 1 & 3 \\ 0 & 2 & -6 \\ 0 & 0 & -8 \end{pmatrix} \begin{bmatrix} R_3' = R_3 - R_2 \end{bmatrix}$$

③ Find the echelon form of $\begin{pmatrix} 2 & 1 & -1 & 2 \\ -1 & 3 & 2 & 3 \\ 1 & 2 & 1 & -1 \\ 2 & -3 & -1 & 4 \end{pmatrix}$

Sol^n
Reduce the matrix to row echelon form by means of elementary row operations:

$$\sim \begin{pmatrix} 2 & 1 & -1 & 2 \\ 0 & 1 & 3 & 8 \\ 0 & 5 & 3 & 2 \\ 0 & -4 & -7 & 6 \end{pmatrix} \begin{bmatrix} R_4' = R_4 + 2R_3 \\ R_3' = R_3 + R_2 \\ R_2' = 2R_2 + R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 2 & 1 & -1 & 2 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & -12 & -38 \\ 0 & 0 & 6 & 44 \end{pmatrix} \begin{bmatrix} R_4' = 5R_4 + 7R_3 \\ R_3' = R_3 - 5R_2 \end{bmatrix}$$

$$\sim \begin{pmatrix} 2 & 1 & -1 & 2 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & -12 & -38 \\ 0 & 0 & 6 & 50 \end{pmatrix} \begin{bmatrix} R_4' = 2 \times R_4 + R_3 \end{bmatrix}$$

30 August 2026

Find the reduced row echelon form of $\begin{pmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{pmatrix}$

Soln:-

Reduce the matrix to reduced row echelon form by means of elementary operation.

$$\sim \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{pmatrix} \left[R_2' = \frac{R_2}{2}\right]$$

$$\sim \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{pmatrix} \left[R_3' = R_3 - 3R_2\right]$$

$$\sim \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{pmatrix} \left[R_3' = R_3(-2)\right]$$

$$\sim \begin{pmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{pmatrix} \left[R_1' = (-1)R_2 + R_1\right]$$

$$\sim \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{pmatrix} \begin{bmatrix} R_1' = \frac{-11}{2}R_3 + R_1 \\ R_2' = \frac{7}{2}R_3 + R_2 \end{bmatrix}$$

Which is the reduced row echelon matrix

(*) Find the reduced row echelon form of

$$\begin{pmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & 5 & 6 & -5 & -1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{pmatrix} \quad [R_1 \leftrightarrow R_2]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{pmatrix} \quad \left[R_1' = \frac{R_1}{2}\right]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{pmatrix} \quad [R_3' = R_3 - 2R_1]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{pmatrix} \quad \left[R_2' = R_2\left(-\frac{1}{2}\right)\right]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 1 \end{pmatrix} \quad [R_3' = R_3 - 5R_2]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad [R_3' = R_3 \cdot 2]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad [R_2' = \frac{7}{2}R_3 + R_2]$$

$$\sim \begin{pmatrix} 1 & 2 & -5 & 3 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad [R_1' = -6R_3 + R_1]$$

$$\sim \begin{pmatrix} 1 & 2 & 0 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad [R_1' = 5R_2 + R_1]$$

CT ৪০ ২য়ের প্রমাণ
একটা Algebric Plane আর একটা Matrix নেবো

* Find the reduced row echelon form of

$$\begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & -2 & 0 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{pmatrix} \quad [R_2' = R_2 - 2R_1,\ R_4' = R_4 - 2R_1]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{pmatrix} \quad [R_2' = (-1)R_2]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 \end{pmatrix} \quad R_3 \quad [R_3' = R_3 - 5R_2,\ R_4' = R_4 - 4R_2]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_4' = R_4 \times \tfrac{1}{6}]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_3 \longleftrightarrow R_4]$$

$$\sim \begin{pmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_2' = -3R_3 + R_2]$$

$$\sim \begin{pmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad [R_1' = 2R_2 + R_1]$$

1/9/26

Find the reduced row echelon form of

① $\begin{pmatrix} 3 & 4 & -1 & 2 \\ 1 & -2 & 3 & 1 \\ 3 & 14 & -11 & 1 \end{pmatrix}$ ② $\begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 4 \\ 0 & 7 & 10 \end{pmatrix}$

① $\left(\begin{array}{cccc} 3 & 4 & -1 & 2 \\ 1 & -2 & 3 & 1 \\ 3 & 14 & -11 & 1 \end{array}\right)$

$\sim \left(\begin{array}{cccc} 1 & -2 & 3 & 1 \\ 3 & 4 & -1 & 2 \\ 3 & 14 & -11 & 1 \end{array}\right)$ $[R_1 \longleftrightarrow R_2]$

$\sim \left(\begin{array}{cccc} 1 & -2 & 3 & 1 \\ 0 & 10 & -10 & -1 \\ 0 & 20 & -20 & -2 \end{array}\right)$ $\begin{bmatrix} R_2' = R_2 - 3R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$

$\sim \left(\begin{array}{cccc} 1 & -2 & 3 & 1 \\ 0 & 1 & -1 & -\frac{1}{10} \\ 0 & 20 & -20 & -2 \end{array}\right)$ $\left[R_2' = \frac{R_2}{10}\right]$

$\sim \left(\begin{array}{cccc} 1 & -2 & 3 & 1 \\ 0 & 1 & -1 & -\frac{1}{10} \\ 0 & 0 & 0 & 0 \end{array}\right)$ $[R_3' = R_3 - 20R_2]$

$\sim \left(\begin{array}{cccc} 1 & 0 & 1 & \frac{9}{5} \\ 0 & 1 & -1 & -\frac{1}{10} \\ 0 & 0 & 0 & 0 \end{array}\right)$ $[R_1' = R_1 + 2R_2]$

(2) $\begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 1 \\ 0 & 7 & 10 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 10 \\ 0 & 7 & 10 \end{pmatrix}$ $[R_2' = R_2 + 2R_1]$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & \frac{10}{7} \\ 0 & 7 & 10 \end{pmatrix}$ $\left[R_2' = \frac{R_2}{7}\right]$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & \frac{10}{7} \\ 0 & 0 & 0 \end{pmatrix}$ $[R_3' = R_3 - 7R_2]$

$\sim \begin{pmatrix} 1 & 0 & \frac{1}{7} \\ 0 & 1 & \frac{10}{7} \\ 0 & 0 & 0 \end{pmatrix}$ $[R_1' = R_1 - 2R_2]$

(3) $\begin{pmatrix} 1 & 2 & -3 & -2 & -3 \\ 3 & 8 & -2 & 0 & -4 \\ 1 & 3 & -7 & -2 & -11 \\ 2 & 1 & -9 & -10 & -3 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 2 & -3 & -2 & -3 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 2 & 2 & 4 & -2 \\ 0 & -3 & -3 & -6 & 3 \end{pmatrix}$ $\begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - 3R_1 \\ R_4' = R_4 - 2R_1 \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 2 & -3 & -2 & -3 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ $\begin{bmatrix} R_3' = R_3 - 2R_2 \\ R_4' = R_4 + 3R_2 \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 0 & -5 & -6 & -1 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ $[R_1' = R_1 - 2R_2]$

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~~Using Row operations~~ Finding inverse matrix

For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 3 & 0 & 8 \end{pmatrix}$, the computations are as follows:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 3 & 0 & 8 & 0 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right) \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right) \begin{bmatrix} R_3 = R_3 + 2R_2 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \begin{bmatrix} R_3' = (-1) R_3 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \begin{bmatrix} R_1' = (-3) R_3 + R_1 \end{bmatrix}$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right) \begin{bmatrix} R_1' = (-2) R_2 + R_1 \end{bmatrix}$$

Thus

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

Conference 03-03-2024 14.27.pdf

1. Find the inverse of the following matrix by elementary operations.

$$\begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$$

Sol^n:-

$$(A|I) = \left(\begin{array}{ccc|ccc} 3 & -3 & 4 & 1 & 0 & 0 \\ 2 & -3 & 4 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & \frac{4}{3} & \frac{1}{3} & 0 & 0 \\ 2 & -3 & 4/3 & -2/3 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \left[R_1' = \frac{R_1}{3}\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 4/3 & 1/3 & 0 & 0 \\ 0 & -1 & 4/3 & -2/3 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \left[R_2' = R_2 - 2R_1\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 4/3 & 1/3 & 0 & 0 \\ 0 & 1 & -4/3 & 2/3 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \left[R_2' = -R_2\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 4/3 & 1/3 & 0 & 0 \\ 0 & 1 & -4/3 & 2/3 & -1 & 0 \\ 0 & 0 & -1/3 & 2/3 & -1 & 1 \end{array}\right) \left[R_3' = R_3 + R_2\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 4/3 & 1/3 & 0 & 0 \\ 0 & 1 & -4/3 & 2/3 & -1 & 0 \\ 0 & 0 & 1 & -2 & 3 & -3 \end{array}\right) \left[R_3' = -3R_3\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 3 & -4 & 4 \\ 0 & 1 & 0 & -2 & 3 & -4 \\ 0 & 0 & 1 & -2 & 3 & -3 \end{array}\right) \left[\begin{array}{l} R_1' = -\frac{4}{3}R_3 + R_1 \\ R_2' = \frac{4}{3}R_3 + R_2 \end{array}\right]$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -2 & 3 & -4 \\ 0 & 0 & 1 & -2 & 3 & -3 \end{array}\right) \left[R_1' = R_2 + R_1\right]$$

$$\sim (I \mid A^{-1})$$

Thus $A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{pmatrix}$

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Spring 2026

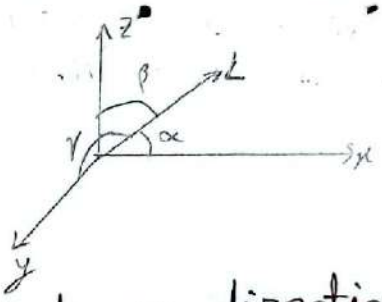
Class Test 1(Section A)	Program: B.Sc in CSE
Registration No.: 25101003	Course Title: Math III
Name: Utsob Ghosh	Course Code: Math 201
Time:30 minutes	Total Marks:10

1. Define Direction Cosine of a line. Find the relationship between Direction Cosines.
2. Find the perpendicular distance from the point (4, -3) to the line $5x + 3y - 11 = 0$.

Answer to the question no -1

If a line OL makes angle α, β, γ with axes x-axes, y-axes, z-axes respectively then $\cos\alpha$, $\cos\beta$, $\cos\gamma$ are called direction cosines. They are denoted as l, m, n respectively.

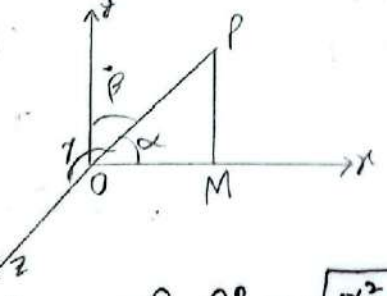
So, direction cosines of OL $= (l, m, n) = (\cos\alpha, \cos\beta, \cos\gamma)$



Relationship between direction cosines

Let, a line OP make angle α, β, γ with x-axes, y-axes, z axes respectively. Direction cosines are l, m, n.

$l = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$



Position vector of $OP = \sqrt{x^2+y^2+z^2}$

$\Rightarrow OP^2 = x^2+y^2+z^2 \ldots\ldots (1)$

Draw a perpendicular PM on x-axes ~~on x-axes~~ from P.

From ΔOPM, $\cos\alpha = \frac{PM}{OP} = \frac{x}{OP}$

$\Rightarrow x = OP\cos\alpha$

$\Rightarrow x = OP \times l$

Similarly, $y = OP \times m$ and $z = OP \times n$

$\therefore x^2+y^2+z^2 = (OP\times l)^2+(OP\times m)^2+(OP\times n)^2 = OP^2l^2+OP^2m^2+OP^2n^2$

$\Rightarrow x^2+y^2+z^2 = OP^2(l^2+m^2+n^2)$

$\Rightarrow OP^2 = OP^2(l^2+m^2+n^2)$ [From equation ①]

$\Rightarrow \frac{OP^2}{OP^2} = l^2+m^2+n^2$

$\Rightarrow 1 = l^2+m^2+n^2$

$\therefore l^2+m^2+n^2 = 1$

It is the relationship between direction cosines.

Answer to the question no-2

We know, If a line is $ax+by+c$ then the perpendicular line of the line is $bx-ay+k=0$

Given,

$5x+3y-11=0$

here, $a=5, b=3, c=-11$

∴ perpendicular line is $3x-5y+k=0 \ldots$ ①

Perpendicular distance from the point $(4,-3)$ to the line $5x+3y-11=0$ is

$$\frac{5\times 4+3\times(-3)-11}{\sqrt{5^2+3^2}}$$

$$= \frac{20+(-9)-11}{\sqrt{25+9}}$$

$$= \frac{0}{\sqrt{34}}$$

$$= 0$$

Since the point $(4,-3)$ is located in the line $5x+3y-11=0$ the perpendicular distance is 0.

Spring 2026

Class Test 2(Section A)	Program: B.Sc in CSE
Registration No.: 25101003	Course Title: Math III
Name: Utsob Ghosh	Course Code: Math 201
Time:20 minutes	Total Marks:10

1. Show that the four points (-6, 3, 2), (3, -2, 4), (5, 7, 3), (-13, 17, -1) are coplanar.
2. Find the reduced row Echelon form of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$.

1. The four points (-6, 3, 2), (3, -2, 4), (5, 7, 3), (-13, 17, -1) are co-planar if their determinant is zero.

Now,

$$\begin{vmatrix} -6 & 3 & 2 & 1 \\ 3 & -2 & 4 & 1 \\ 5 & 7 & 3 & 1 \\ -13 & 17 & -1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -6 & 3 & 2 & 1 \\ 9 & -5 & 2 & 0 \\ 11 & 4 & 1 & 0 \\ -7 & 14 & -3 & 0 \end{vmatrix} \quad \begin{bmatrix} R_4' = R_4 - R_1 \\ R_3' = R_3 - R_1 \\ R_2' = R_2 - R_1 \end{bmatrix}$$

$$= -1 \times \begin{vmatrix} 9 & -5 & 2 \\ 11 & 4 & 1 \\ -7 & 14 & -3 \end{vmatrix} + 0 - 0 + 0$$

$$= -1 \times \begin{vmatrix} 9 & 0 & 0 \\ 11 & 91 & -13 \\ -7 & 91 & -13 \end{vmatrix} \quad \begin{bmatrix} C_2' = 9C_2 + 5C_1 \\ C_3' = 9C_3 - 2C_1 \end{bmatrix}$$

$$= -1 \times 9 \times \{91 \times (-13) - 91 \times (-13)\} = -9(-1183 + 1183)$$

$$= -9 \times 0$$

$$= 0$$

Since their determinant is zero the four points are coplanar

2. $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & -2 & 5 \end{pmatrix}$ $\begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{pmatrix}$ $\begin{bmatrix} R_3' = R_3 + 2R_2 \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 0 & 9 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{pmatrix}$ $\begin{bmatrix} R_1' = R_1 - 2R_2 \end{bmatrix}$

~~Which is the required reduced row echelon form of the matrix~~

$\sim \begin{pmatrix} 1 & 0 & 9 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}$ $\begin{bmatrix} R_3' = \frac{R_3}{-1} \end{bmatrix}$

$\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $\begin{bmatrix} R_2' = R_2 + 3R_3 \\ R_1' = R_1 - 9R_3 \end{bmatrix}$

Which is the ~~requed~~ required reduced row echelon ~~matrix~~ form of the matrix.